

Shubham Pande

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RESEARCH INTEREST

Advanced CMOS Devices, Novel Memory Concepts, Neuromorphic Hardware, Design–Technology Co-Optimization (DTCO).

PROFESSIONAL EXPERIENCE

Intel Technology, India Pvt. Ltd.

Device Modeling Engineer

June 2024 – Present

Hyderabad, India

- Building PDKs for nanosheet CMOS devices with backside power delivery to support early IP validation and accelerate design bring-up on 14A/10A nodes.
- Standardizing CMOS device and interconnect models for EDA integration.
- Partnering with TD/Design Rules/Std-Cell/Circuit teams to align model targets with product performance & reliability requirements (DTCO).

EDUCATION

Indian Institute of Technology, Madras

Direct PhD (MS + PhD)

2019 – 2025

CGPA - 9.41

SGGSIE&T, Vishnupuri, Nanded, Maharashtra

B.Tech in Electronics and telecommunication

2012 – 2016

CGPA - 8.02

RESEARCH EXPERIENCE

PhD Research

2020 – 2025

Advisors: Prof. Anjan Chakravorty & Prof. Bhaswar Chakrabarti

IIT Madras, India

- Thesis Title: ReRAM Memory Technology for Neuromorphic Computing – Design, Modelling, and Applications.
- Developed physics-based compact models and performed electrical characterization of ReRAM devices for analog compute-in-memory.
- Quantified electrothermal effects in ReRAM crossbar arrays, showing their impact on energy efficiency and reliability.
- Exploiting the electrothermal effects in ReRAM, proposed low-power ReRAM cell design.

Fulbright-Nehru Doctoral Research Fellow, UC Santa Barbara

Advisor: Prof. Dmitri Strukov

Aug 2022 – July 2023

UC Santa Barbara, USA

- Designed temperature-modulated synaptic devices for online learning in spiking neural networks (SNNs).
- Bridged device-level physics and neural architectures, enhancing interdisciplinary expertise.

MS Research

2019 – 2020

Advisor: Prof. Anjan Chakravorty

IIT Madras, India

- Developed compact models for thermal resistance and transient heating in SiGe HBTs, validated experimentally.
- Proposed methods to mitigate self-heating, improving device reliability.
- Gained expertise in multiphysics modeling (thermal–electrical interactions).

Development of Low-Cost BiCMOS Process

2018 – 2019

Advisors: Prof. Anjan Chakravorty & Prof. Nihar Mohapatra (IITGN, India)

IIT Madras, India

- Co-developed a cost-effective BiCMOS platform by enhancing RF/DC performance of parasitic BJTs in 180 nm CMOS.
- Developed expertise in device simulation, co-optimization, and RF layout design, parameter extraction.

Research Assistant

2017 – 2018

Advisors: Prof. Nihar Mohapatra

IITGN, India

- Worked on developing cost-effective integration of 20–40V n/p LDMOS devices in SCL's 0.18 μ m CMOS process.

CORE SKILLS

Simulation/Model. Synopsys Sentaurus TCAD (Sprocess, SDE, Sdevice), COMSOL Multiphysics.

EDA/Analysis : Cadence - Spectre/Virtuoso, QucsStudio, LTspice, Klayout.

Prog. languages : Verilog-A, Python (Libraries used: Matplotlib, NumPy, SciPy, Pandas, Scikit-learn), LaTeX.

Domains : Compute-in-memory, Image sensor pixels (FeFET/ReRAM-coupled), Reliability, SNNs.

PATENTS

- **Shubham Pande**, B. Chakrabarti, and A. Chakravorty, "A ReRAM device for low power operation and a method for fabrication thereof", Indian patent granted, patent no: 566655.
- **Shubham Pande**, B. Chakrabarti, and A. Chakravorty, "Apparatus for Storing Information on a Semiconductor Device, and Method Thereof", Indian patent filed, application no. 202341055188, 17 Aug 2023.

PUBLICATIONS

Journal publications

- * **Shubham Pande**, T. Bhattacharya, S.S. Bezugam, E. Wlazlak, and D. Strukov, "Accelerating Neuromorphic Hardware Training with ReRAM NeoHebbian Synapses", Under review.
- * **Shubham Pande**, S. Balanethiram, B. Chakrabarti, and A. Chakravorty, "A Physics-Based Compact Model for Thermal Resistance in Resistive Random-Access Memories", Solid-State Electronics, vol. 204, 2023.
- * Masud R. Sk, **Shubham Pande**, F. Müller, Y. Raffel, M. Lederer, L. Pirro, S. Beyer, K. Seidel, T. Kämpfe, S. De, and B. Chakrabarti, "Fixed Charges at the HfO₂/SiO₂ Interface: Impact on The Memory Window of FeFET", Memories - Materials, Devices, Circuits and Systems, Vol. 4, 2023.
- * L.H. Reddy, **Shubham Pande**, T. Roy, E.M. Vogel, A. Chakravorty, and B. Chakrabarti, "A SPICE compact model for forming-free, low-power graphene-insulator-graphene ReRAM technology", Emergent Materials, 4, 1055-1065, 2021.
- * Nidhin K., **Shubham Pande**, S. Yadav, S. Balanethiram, D. R. Nair, S. Frégonèse, T. Zimmer, and A. Chakravorty, "An Efficient Thermal Model for Multifinger SiGe HBTs Under Real Operating Conditions" IEEE Transactions on Electron Devices 67, 2020.

Conference publications (selected)

- * T. Bhattacharya*, S.S. Bezugam*, **Shubham Pande***, E. Wlazlak, and D. Strukov, "ReRAM-Based NeoHebbian Synapses for Faster Training-Time-to-Accuracy Neuromorphic Hardware", 2023 International Electron Devices Meeting (IEDM), 2023, pp. 1-4, *Equal contribution.
- * **Shubham Pande**, S. Chourasia, and B. Chakrabarti, "3T-1C-1R Synaptic Design for Online Learning in Spiking Neural Networks", IEEE Device Research Conference (DRC), 2025.
- * S. Chourasia, **Shubham Pande**, Andrea Padovani, and B. Chakrabarti, "Leveraging Capacitance Modulation of ReRAM for CMOS-ReRAM Image Sensor", IEEE Device Research Conference (DRC), 2025.
- * **Shubham Pande**, M.R. Sk, Y. Raffel, M. Lederer, K. Seidel, A. Chakravorty, S. De, and B. Chakrabarti, "FeFET based LIF Neuron with Learnable Threshold and Time Constant", IEEE Device Research Conference (DRC), 2024.
- * **Shubham Pande**, B. Chakrabarti, and A. Chakravorty, "On the Prevalence of Row Hammer Attacks in FeFETs based Memory Systems", IEEE Electron Device Technology Meeting (EDTM), 2024.
- * **Shubham Pande**, B. Chakrabarti, and A. Chakravorty, "Electro-thermal Confinement Leads to Low Power ReRAM Operation", XXII International Workshop on Physics of Semiconductor Devices, 2023.
- * **Shubham Pande**, B. Chakrabarti and A. Chakravorty, "Leveraging MOSFET V_{th} Temperature Sensitivity for Memory Application", Non-volatile Memory Technology Symposium (NVMTS), imec, Belgium, 2023.
- * **Shubham Pande**, S. Balanethiram, B. Chakrabarti, and A. Chakravorty, "2D-Motion Detection using SNNs with Graphene-Insulator-Graphene Memristive Synapses", 29th IEEE International Conference on Electronics, Circuits, and Systems (ICECS), 2022.
- * **Shubham Pande**, Nidhin K., S. Yadav, S. Balanethiram, and A. Chakravorty, "Thermal Impedance Model for Multifinger SiGe HBTs", IEEE Electron Device Technology Meeting (EDTM), 2024.
- * **Shubham Pande**, S. Balanethiram, A. K. Singh, M. Gupta, B. Umapathi, H. S. Jatana, N. R. Mohapatra, and A. Chakravorty, "Development of Low-Cost Silicon BJT Technology and Modeling", 5th IEEE International Conference on Emerging Electronics (ICEE), 2020.
- * S. Balanethiram, **Shubham Pande**, A. K. Singh, B. Umapathi, H. S. Jatana, N. R. Mohapatra, and A. Chakravorty, "Development of Low-Cost Silicon BiCMOS Technology for RF Applications", IEEE Conference on Modeling of Systems Circuits and Devices (MOS-AK India), 2019.

BOOK CHAPTERS

- B. Chakrabarti, **Shubham Pande**, S. Roy, and M. Sk. Rana, "Oxygen vacancy based resistive memory for storage and computing applications", in press.
- Sourav De, **Shubham Pande**, Masud Rana Sk, Ankush Chattopadhyay, Yannick Raffel, B. Chakrabarti, and Tuo-hung Hou, "Hafnium Oxide-Based Ferroelectric Memories: Applications and Future Prospects", in press.

FELLOWSHIPS

- Fulbright-Nehru Doctoral Research Fellowship - Aug 2022 – July 2023
- Prime Minister's Research Fellowship (PMRF) 2020 - 2024
- Central Sector Scholarship, 2012 - 2016